

Asterisk is open-source software that can transform an ordinary PC into an IP PBX system. It has the versatility to power a wide array of telephony applications, including call centers, internet telephony services, and any custom telephony solution you can imagine. According to asterisk.org, there are over **1 million Asterisk servers** operating worldwide, making it one of the most significant projects in the world of telecommunication.

The Origins of the Asterisk Project

In the early days, IP telephony was costly. Businesses paid around \$1,500 for a voice logger, \$2,500 for interactive voice response, and \$4,000 for call distribution and computer telephony integration systems. When Mark Spencer, founder of "Linux Support Services," needed a telephone system for his company in 1999, the quotes he received were over \$50,000—well beyond budget.

Spencer, leveraging his telecommunications expertise, decided to develop his own telephony software. Within months, he had created a functional system, which he shared publicly under a GPL license. By 2001, **Linux Support Services** was rebranded as **Digium**, and with contributions from developers worldwide, Asterisk continued to evolve as a **free, open-source, and disruptive telephony solution**.



Mark Spencer - Founder Asterisk

How to Build Your Own PBX with Asterisk

Let's look at the components needed to set up an Asterisk IP PBX and the big picture:

1. **Hardware Selection**
 2. **Software Installation**
 3. **Trunk Line Connections**
 4. **Phone Options**
 5. **Configuration**
-

1. Hardware

Your server's specifications depend on expected call volumes and integrations with third-party applications. For a 500-user PBX, a stable, always-on server is essential to handle 24/7 operations.



Recommended Server Specifications for 500 users IP PBX with 120 concurrent calls:

Processor - Intel Core i5 or above

Memory - 4/8 GB

Disk - 500 GB/1TB HDD

Asterisk Server

- **Supported OS:** Asterisk is compatible with Fedora, Ubuntu, CentOS, RHEL, and Debian Linux.
- **Configuration:** A normal PC with moderate specifications can handle smaller setups, though stable hardware is recommended for larger, enterprise-level deployments.

2. Software

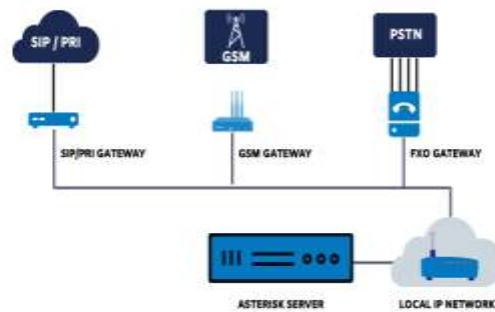
The simplest way to set up Asterisk is to download and install the software directly. This approach is highly flexible but requires manual configuration in the backend. Asterisk doesn't include a GUI by default, so I recommend learning Asterisk fundamentals for a smooth setup.



Asterisk - FreePBX - Issabel PBX - VoIP Savvy

Alternatively, you can opt for **FreePBX** or **Issabel**, which provide a graphical interface and easier configuration options. **Issabel** is a continuation of **Elastix**, which offers an enhanced GUI and an excellent contact center interface.

3. Connecting Trunk Lines



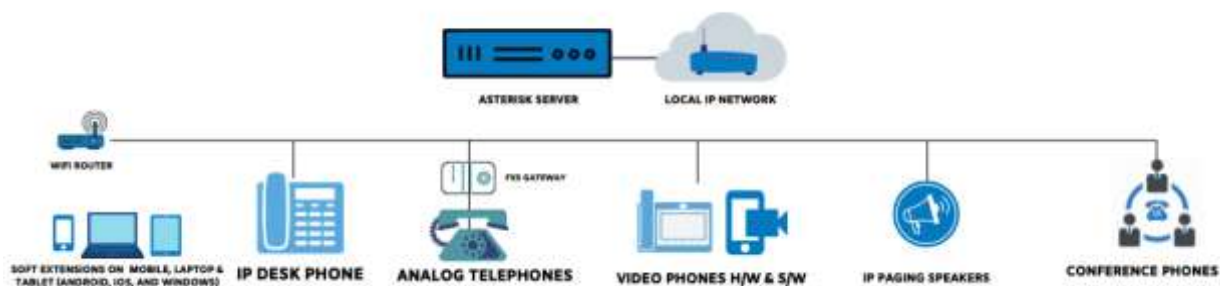
VoIP Savvy Asterisk

Asterisk can connect to public networks through various trunk lines:

- **Analog PSTN:** Connect using an FXO gateway.
- **E1/T1:** Use an E1 card or gateway to link E1 PRI lines.
- **SIP Trunk:** No extra hardware needed—simply connect via the server's Ethernet port.
- **GSM Network:** Connect with a GSM gateway, a cost-effective solution in regions where mobile rates are lower.

4. Phone Options

Asterisk connects seamlessly with any **SIP-based extension**. Here's a quick look at compatible options:



Asterisk VoIP Savvy Telephone Extensions

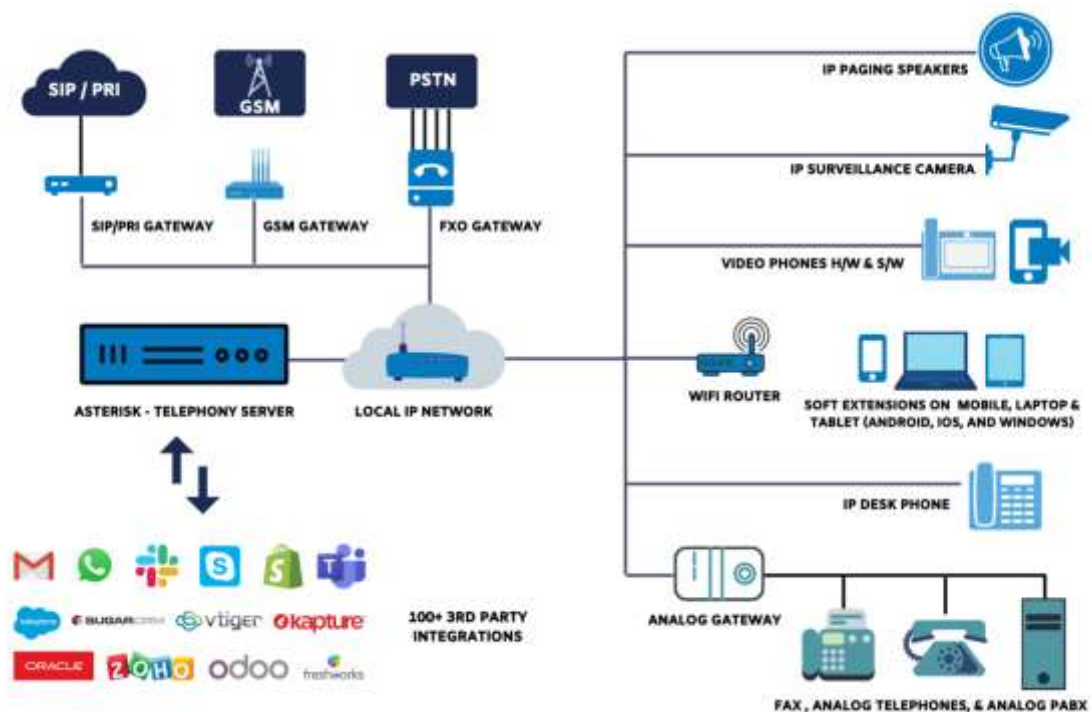
- **Softphones:** Free apps for desktop and mobile (Android/iOS) that pair well with Asterisk.
- **IP Desk Phones:** Available from Yealink, Grandstream, Sangoma, Polycom, and Fanvil, with prices starting around \$30.
- **Analog Phones:** Connect existing analog phones using an **FXS gateway**.
- **Additional Devices:** Conference phones, SIP speakers (for paging), and video phones can all be integrated.

5. Configuration

There are numerous online resources and step-by-step guides for setting up Asterisk. If you're tech-savvy, you can configure it yourself from scratch. For professional setup, **VoIP Savvy** offers **Asterisk training, consultation, installation, configuration, third-party integrations, and 24/7 technical support.**

The Big Picture: Asterisk IP PBX Architecture

Here's what a complete Asterisk IP PBX setup looks like:



VoIP Savvy Asterisk IPPBX Architecture

1. **Trunk Lines:** Compatible with GSM, E1 PRI, SIP, VoIP, and PSTN, allowing for flexible network configurations based on your business needs.
2. **Extensions:** Supports softphones, IP phones, analog phones, conference room phones, and paging speakers.
3. **Network Location:** Deploy Asterisk on a local server or in the cloud; integrate it with your enterprise apps through APIs.

Asterisk's flexibility allows it to serve a variety of functions, from IP PBX and call centers to interactive voice response and web telephony (WebRTC).

Getting started

- 1.To begin, we need to install Asterisk from terminal with apt. It is advisable to keep apt package installer up to date by issuing the `sudo apt update` command.
- 2.Install Asterisk with the command `sudo apt install asterisk-dahdi`

```
thaprinced@thaprinced-Aspire-E5-571:~$ sudo apt install asterisk-dahdi
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

The following additional packages will be installed:
  asterisk asterisk-config asterisk-core-sounds-en asterisk-core-sounds-en-gsm
  asterisk-modules asterisk-moh-opsound-gsm dahdi dahdi-dkms dahdi-linux
  freetds-common fxload gawk libc-client2007e libcodec2-1.0 libgmime-3.0-0
  libgsm1 libiksemel3 liblua5.2-0 libneon27 libodbc2 libopencore-amrnb0
  libopencore-amrwb0 libopenr2-3 libopusfile0 libportaudio2 libpq5 libpri1.4
  libradcli4 libresample1 libsigsegv2 libsox-fmt-alsa libsox-fmt-base libsox3
  libspandsp2 libsrtp2-1 libss7-2.0 libsybdb5 libtonezone2.0 libunbound8
  liburiparser1 libvo-amrwbenc0 mlock sox
Suggested packages:
  asterisk-dev asterisk-doc asterisk-ooh323 asterisk-opus asterisk-vpb
  gawk-doc uw-mailutils odbc-postgresql tdsodbc libsox-fmt-all
The following NEW packages will be installed:
  asterisk asterisk-config asterisk-core-sounds-en asterisk-core-sounds-en-gsm
  asterisk-dahdi asterisk-modules asterisk-moh-opsound-gsm dahdi dahdi-dkms
  dahdi-linux freetds-common fxload gawk libc-client2007e libcodec2-1.0
  libgmime-3.0-0 libgsm1 libiksemel3 liblua5.2-0 libneon27 libodbc2
  libopencore-amrnb0 libopencore-amrwb0 libopenr2-3 libopusfile0 libportaudio2
  libpq5 libpri1.4 libradcli4 libresample1 libsigsegv2 libsox-fmt-alsa
  libsox-fmt-base libsox3 libspandsp2 libsrtp2-1 libss7-2.0 libsybdb5
  libtonezone2.0 libunbound8 liburiparser1 libvo-amrwbenc0 mlock sox
```

3. After installation, we can have a look at the content of the package directory by issuing the command `ls /etc/asterisk`. There are several files in the package directory, the most important ones for us is the `sip.conf` and `extensions.conf` files.

```
thapracegthaprace-Aspire-E5-571: $ ls /etc/asterisk
acl.conf               confbridge.conf       pjsip_notify.conf
adsl.conf              config_test.conf      pjsip_wizard.conf
agents.conf            console.conf           prometheus.conf
alarmreceiver.conf     dbsep.conf            queuerules.conf
alsa.conf              dnsmgr.conf           queues.conf
amd.conf               dsp.conf              res_config_mysql.conf
app_mysql.conf         enum.conf             res_config_sqlite3.conf
app_skel.conf          extconfig.conf        res_config_sqlite.conf
ari.conf               extensions.ael        res_corosync.conf
ari_debug_tools.conf  extensions.conf       res_curl.conf
asterisk.adst          extensions.lua         res_fax.conf
asterisk.conf          extensions_minlnv.conf res_ldap.conf
calendar.conf         features.conf         res_odbcc.conf
ccss.conf             festival.conf         resolver_unbound.conf
cdr_adaptive_odbc.conf followme.conf         res_parking.conf
cdr_asterisk.conf     func_odbcc.conf      res_pgsql.conf
cdr.conf              hep.conf              res_pktccops.conf
cdr_custom.conf       http.conf             res_snmp.conf
cdr_manager.conf      iax.conf              res_stun_monitor.conf
cdr_mysql.conf        iaxprov.conf         rp.conf
cdr_odbcc.conf        indications.conf     say.conf
cdr_pgsql.conf        logger.conf          sip.conf
cdr_sqlite3_custom.conf manager.conf          sip_notify.conf
cdr_syslog.conf       memcache.conf        sla.conf
cdr_tds.conf          meetme.conf          smdl.conf
cel_asterisk.conf     minlnv.conf          sorcery.conf
cel.conf              nlsda.conf           ss7_timers
cel_custom.conf       modules.conf         status.conf
cel_odbcc.conf        notify.conf          statusd.conf
cel_pgsql.conf        musiconhold.conf     stir_shaken.conf
cel_sqlite3_custom.conf muted.conf            telcordia-1.adst
cel_tds.conf          ooh323.conf          test_sorcery.conf
chan_dahdi.conf       osp.conf             udptl.conf
chan_mobile.conf      oss.conf             users.conf
cli_aliases.conf      phone.conf           voicemail.conf
cli.conf              phoneprov.conf       vpb.conf
cli_permissions.conf  pjsip.conf           xmpp.conf
```

Server Configuration

We will now configure our server so we can connect our telephone extensions. We will do this by writing scripts to the files mentioned earlier; sip.conf and extensions.conf.

1. Since we will be altering these two files, it is best practice to make copies of them in case we need to roll back on configurations in the future. Use the command `sudo cp sip.conf sip.conf.backup` and `sudo cp extensions.conf.backup` to back up the two files respectively.
2. Issue the command `echo | sudo tee sip.conf` to clear all existing code lines in the sip.conf file
3. Open the sip.conf file with nano text editor using the command `sudo nano sip.conf`
4. Then write the following configuration lines into the sip.conf file:

[general]

context=default

allowoverlap=no

udpbindaddr=0.0.0.0

tcpenable=no

tcpbindaddr=0.0.0.0

transport=udp

srvlookup=yes

[9090]

type=friend

host=dynamic

secret=9090

[9091]

type=friend

host=dynamic

secret=9091

```
[general]
context=default
allowoverlap=no
udpbindaddr=0.0.0.0
tcpnableno
tcpbindaddr=0.0.0.0
transport=udp
priorityjumpings=no

[9090]
type=friend
hostdynamic
secret=9090

[9091]
type=friend
hostdynamic
secret=9091
```

5. Press ctrl + O to save the changes, then ctrl + X to close the file. In our configuration, we have provided a general setup for the server including context, allowoverlap, udpbindaddr, etc for the server. We have also initialized two telephone line extensions (9090, 9091) we will be using later for placing calls. These extensions can be considered as two accounts with dynamic host, meaning their IPs are not static but are dynamically assigned and with passwords provided in the secret field thus 9090 and 9091 respectively.

6. We now move on to the extensions.conf file for further configuration. We repeat steps 2 and 3 replacing 'sip' with 'extensions' accordingly.

7. We then write the following configuration lines into extensions.conf file:

[general]

static=yes

writeprotect=no

priorityjumpings=no

autofallthrough=yes

clearglobalvars=no

[default]

exten => 9090,1,Dial(SIP/9090,10)

exten => 9091,1,Dial(SIP/9091,10)

```

GNU nano 2.2.2 extensions.conf
[general]
static=yes
writeprotect=no
priorityjumps=no
autofallthrough=yes
clearglobalvars=no

[default]
exten => 9090,1,Dial(SIP/9090,10)
exten => 9091,1,Dial(SIP/9091,10)

```

8. Press `ctrl + O` to save the changes, then `ctrl + X` to close the file. We have directly linked the two accounts in our `sip.conf` file with a call action in the `extensions.conf` file. More configuration options can be found on the Asterisk documentation page [here](#).

9. Restart the server with the command `sudo systemctl restart asterisk`. Then we can check if its up and running with the command `sudo systemctl status asterisk`. Finally issue the command `ip a` to get the IP of the server and write it down.

```

thaprince@thaprince-Aspire-E5-571: /etc/asterisk
Active: active (running) since Sat 2022-06-11 10:33:19 GMT; 29s ago
Docs: man:asterisk(8)
Main PID: 22361 (asterisk)
Tasks: 76 (limit: 9367)
Memory: 42.6M
CPU: 1.547s
CGroup: /system.slice/asterisk.service
└─22361 /usr/sbin/asterisk -g -f -p -U asterisk
   22362 astcanary /var/run/asterisk/alt.asterisk.canary.tweet.tweet

Jun 11 10:33:18 thaprince-Aspire-E5-571 systemd[1]: asterisk.service: Deactivat
Jun 11 10:33:18 thaprince-Aspire-E5-571 systemd[1]: Stopped Asterisk PBX.
Jun 11 10:33:18 thaprince-Aspire-E5-571 systemd[1]: asterisk.service: Consumed
Jun 11 10:33:18 thaprince-Aspire-E5-571 systemd[1]: Starting Asterisk PBX...
Jun 11 10:33:19 thaprince-Aspire-E5-571 asterisk[22361]: readctl: rc.read_conf
Jun 11 10:33:19 thaprince-Aspire-E5-571 asterisk[22361]: readctl: rc.read_conf
Jun 11 10:33:19 thaprince-Aspire-E5-571 systemd[1]: Started Asterisk PBX.

thaprince@thaprince-Aspire-E5-571: /etc/asterisk$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 00:7b:1c:72:bd:4a brd ff:ff:ff:ff:ff:ff
21: usb0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 0e:20:ad:03:2f:5e brd ff:ff:ff:ff:ff:ff
    inet 192.168.120.255/24 brd 192.168.120.255 scope global dynamic noprefixroute usb0
        valid_lft 3504sec preferred_lft 3504sec
    inet6 fe80::ede7:b733:b9fa:6235/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
thaprince@thaprince-Aspire-E5-571: /etc/asterisk$

```

Client Configuration

We are now ready to connect our two phones to the asterisk server as clients. To do this, we need to establish a server-client session between the phones and the server over the Session Initiation Protocol mentioned earlier. Thankfully, there are a lot of SIP client apps available on Google play store.

1. For demonstration purposes, download and install two different ones on the two phones. I used Linphone and Mizudroid, you are free to choose any from the play store.

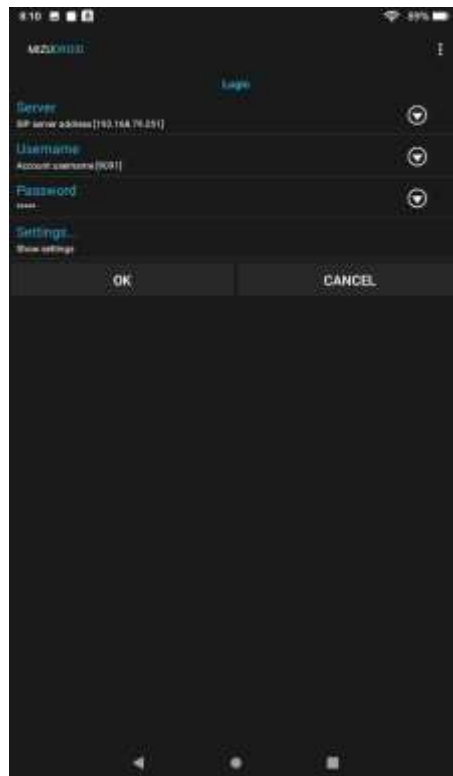
2. After installation, the next thing is to register phones with the two extensions we created on the server. The various apps have different options for connecting your phone to an SIP server, but select either or SIP Account or Basic Account depending on the app you downloaded.

3. Fill out the registration form with the following details:

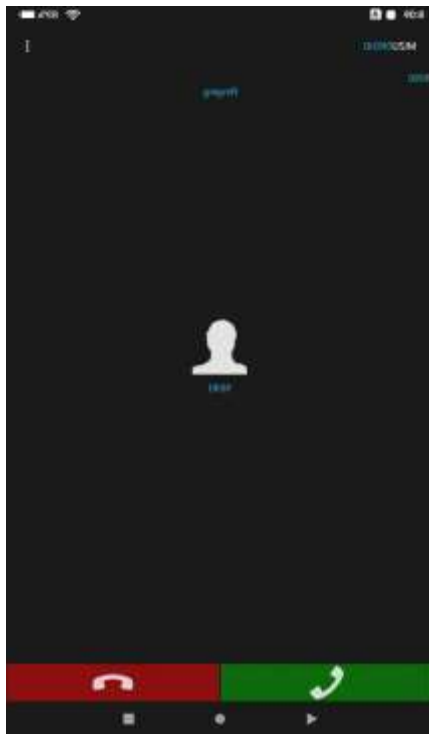
Server/Domain address: enter the server IP address you wrote down earlier.

Username: Use either of the two extensions on the two phone (for example, you can use 9090 on the first phone and 9091 on the second phone).

Password: Enter the value provided in the secret field we provided during the server configuration (for example, 9090 for the 9090 extension and 9091 for the 9091 extension). Then tap register to establish the connection.



4. To verify connectivity, place a call from the first phone to the second with the assigned extension. Cheers if the call goes through, answer the call and have fun with it. In case of any challenges, there are numerous resources online for assistance or get in touch with me.



NB: the computer and the two phones should be on the same LAN (wired or wireless).

To Summarize, the Asterisk application is very open and very extensible.

Asterisk, created by Mark Spencer and Digium (now Sangoma), is a **powerful, free, and open-source** IP PBX solution that continues to evolve and disrupt the telecommunications industry. If you're considering an Asterisk-based PBX for your business, VoIP Savvy can help you understand its integration capabilities and bring your setup to life.

References

Asterisk documentation - <https://wiki.asterisk.org/wiki/display/AST/Hom>